

Trajectory Visualization by Using Global Positioning Systems (GPS)

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Abstract

The purpose of this research is to use GPS to visualize the trajectory of space, air, sea, and land vehicles. This research will reveal the visualization trajectories of two scenarios. One of the scenarios will be an airplane trajectory and the other will be a shuttle launch trajectory. In the first scenario, an airplane flew from Cape Canaveral to Vancouver, Canada. In the second scenario, a rocket was launched from Cape Canaveral and landed near the Corpus Christi, Texas. In order to set up these scenarios, we used a GPS Simulator, GPS receiver, lap-top computer, MATLAB software, FUGAWI software and SATELLITE TOOL KIT (STK) software. This research can be used in a classroom/laboratory environment to give students hands on experience in the integration of hardware and software equipment used for GPS applications.

1. Introduction

In this research, two scenarios were done to illustrate trajectory visualization by using GPS. Conversely, MATLAB was used to differentiate and integrate space data. This research involved the use of GPS Simulators, GPS receivers, computers, MATLAB software, FUGAWI software and SATELLITE TOOL KIT (STK) software.

MATLAB Version 6.5 is a high performance software tool for technical computing [1]. This software was used in the research to integrate and differentiate post Shuttle launch data. The data was from the STS103 Shuttle Launch in December 1999. The goal of this part of the research was to examine the accuracy of MATLAB's differentiation and integration capabilities. MATLAB was used to integrate the Shuttle's velocity in order to obtain position. MATLAB was used to differentiate the Shuttle's position in order to get velocity. MATLAB produced errors on the order of 0.0125 % during the test runs.

The CAST -1000 GPS Simulator is a portable satellite simulator [2]. It can simulate 10 GPS satellites simultaneously. It simulates Coarse/Acquisition (C/A) and P codes. It can generate both L1 and L2 frequencies. L1 frequency is 1575.42 MHz and L2 is 1227.60 MHz. It was used in this research to simulate up to 8 satellites on the L1 frequency.

The ASHTECH G12 GPS receiver is a portable hardware device that processes signals from the GPS satellite constellation [3]. This receiver uses twelve discrete parallel channels for C/A code-phase (pseudo-range) measurements and carrier phase measurements on the L1 (1575.42 MHz) band. The receiver was used in this research to provide real-time position, velocity, and time measurements.

SATELLITE TOOL KIT Version 4.3 is a hands-on software tool designed to provide the skills necessary to perform basic aerospace analysis [4]. It can generate paths for a variety of space-, air-, sea-, and land-based objects, such as satellites, aircraft, ships and ground vehicles. It can be used to calculate and visualize a satellite's position and altitude. STK provides 3D animation capabilities and a 2D map background for visualizing the path of these vehicles over time. Ephemeris data from the STS103 's December 1999 Shuttle Launch was loaded into STK from a text file. STK used the data to simulate the trajectory of this Shuttle Launch.

FUGAWI Version 3.0 is a comprehensive program for integrating bit-mapped charts or maps with a GPS receiver [5]. The FUGAWI software is used for visualization of the trajectory of a moving vehicle. The software was used in this research to visualize the ground tracks of an airplane and a rocket.

2. Experimental Work

2.1 Loading Ephemeris Data Into STK

In order to load Ephemeris data into STK, first create a new Scenario by clicking the Scenario icon upon opening STK. Second in the browser menu select the missile icon. The missile object will be created. Click on the missile object so that it will be highlighted in blue. Third click on Properties in the main menu. Fourth select Basics. The Basics menu appears. Fifth under the Propagator drop-down menu select STK External. Sixth browse through the External Ephemeris File to find the STS103 Shuttle Launch data file. Seventh click the apply button. Eighth click the ok button. STK displayed the trajectory of the shuttle launch as orbit tracks in the 3-D visualization window. STK

displayed the trajectory of the shuttle launch as ground tracks in the 2-D map window. The plot of the trajectory is shown in the Results section.

2.2 MATLAB Computations

The STS103 Shuttle was launched in December 1999. The Ephemeris report showed the output data for velocity in meters per second and position in meters and time in seconds. The velocity was integrated by MATLAB in order to retrieve the position. The percent errors were less than 0.0125 %. The position was differentiated by MATLAB in order to get the velocity. The percent errors were less than 0.0125 %. MATLAB is an excellent tool to use in integrating and differentiating space data. MATLAB outputs used in this research for integration and differentiation are shown in the Results Section.

2.3 Trajectories Produced From Using GPS

Two scenarios were created for producing trajectories. The CAST –1000 GPS Simulator, ASHTECH G12 GPS receiver and the FUGAWI 3.0 software were used for these scenarios. An airplane was the vehicle used in the first scenario. A rocket was the vehicle used in the second scenario.

In the first scenario, the aircraft will be stationary for five minutes. Second, the aircraft changed terminal speed to 500 m/s and changed maximum acceleration to 100 m/s^2 . Third, the airplane reached a terminal height of 10,000 m. It will climb at a maximum angle of 40° . It reached a maximum climb acceleration of 80 m/s^2 . Fourth it cruised for 180 minutes. The airplane traveled from Cape Canaveral, Florida to Vancouver, Canada. The latitude setting was 28° and the longitude was -81° . The true initial heading was 290° from the North. The GPS receiver locked onto 7 satellites. The FUGAWI software showed the airplane trajectory. A plot of the airplane trajectory is shown in the Results Section.

In the second scenario, a very fast helicopter was used to simulate the rocket scenario. First the helicopter was stationary for thirty minutes. Second it took off at a terminal velocity of 1000 m/s, at a maximum acceleration of 50 m/s^2 . During the takeoff, the course heading was 90° and the course climb angle was 90° . Third it pitched up at a climb angle of 60° at a maximum acceleration of 100 m/s^2 . Fourth the helicopter reached a terminal speed of 2000 m/s at a maximum acceleration of 100 m/s^2 . Fifth the helicopter cruised for 30 minutes. The GPS receiver locked onto five satellites for the second scenario. The FUGAWI software showed the rocket trajectory. A plot of the rocket trajectory is shown in the Results Section.

3. Results

In Figure 1, the trajectory of the STS103 Shuttle Launch is shown. The STK software produced the trajectory. In this figure STK is executing its animation capabilities with a two-dimensional background of the Earth. The shuttle launch produced ground tracks in a two-dimensional map window and orbit tracks in a three-dimensional visualization window. These windows allow students to study graphically the relationships between objects. The two windows will allow them to manipulate data and objects.

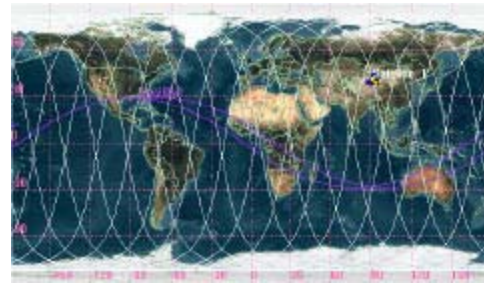


Figure 1: Shuttle Launch Trajectory Produced by STK Software

MATLAB was used to produce the waveform in Figure 2. The waveform represents the velocity trajectory of the Shuttle launch STS103 in orbit. The trajectory was produced by the differentiation of the position vector in the x-direction. This output waveform is the velocity vector in the x-direction. Trajectories representing the y-direction and z-direction for velocity were similar.

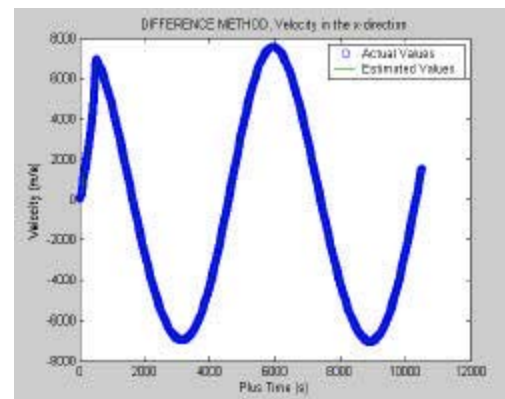


FIGURE 2: Velocity Trajectory Produced by MATLAB

MATLAB was used to produce the waveforms in Figure 3 and Figure 4. The waveform represents the position trajectory of the Shuttle launch STS103 in orbit. The trajectory was produced by the integration of the velocity vector in the y-direction. These output waveforms are the

position vectors in the y-direction. Trajectories representing the x-direction and z-directions for position were similar. MATLAB used the Cumulative Trapezoidal Numerical Integration (CTNI) Method to integrate the velocity data. There is a shift in the two waveforms because CTNI Method started its numerical integration at zero. A MATLAB algorithm was written to include a correction factor (CF) to add to the estimated values of the position vectors. Figure 3 shows the integration without CF. Figure 4 shows the integration with CF.

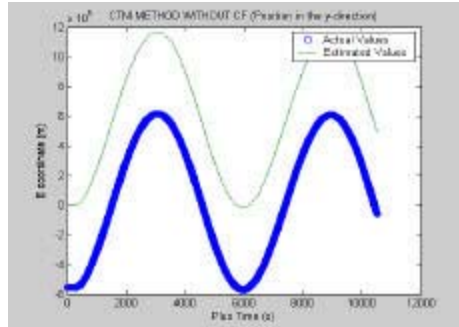


FIGURE 3: Position Trajectory Without CF Produced by MATLAB Algorithm

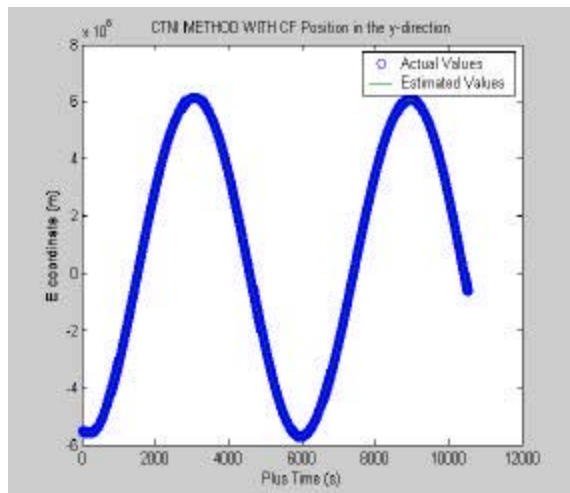


FIGURE 4: Position Trajectory with CF Produced by MATLAB Algorithm

Figure 5 shows the trajectory of the airplane scenario. The airplane started its flight at Cape Canaveral. The final destination was at Vancouver, Canada. The trajectory was created by the integrated use of the GPS Simulator, GPS Receiver and the FUGAWI software.



FIGURE 5: Airplane Trajectory

Figure 6 shows the trajectory of the rocket scenario. The rocket started its launch at Cape Canaveral. The final destination of the rocket was near Corpus Christi, Texas. The trajectory was created by the integrated use of the GPS Simulator, GPS Receiver and the FUGAWI software.

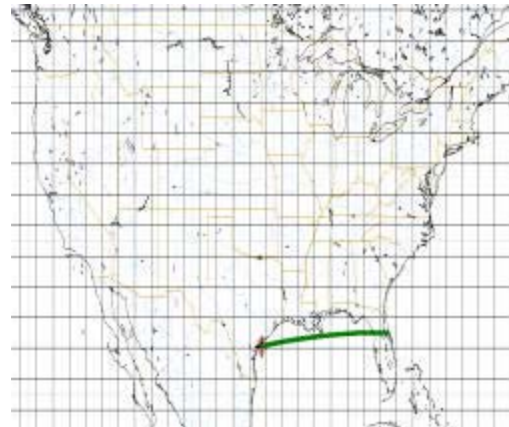


FIGURE 6: Rocket Trajectory

4. Conclusion

In this research, GPS was used for trajectory visualization. In one scenario we created an airplane trajectory. In the second scenario we created a rocket scenario. This research can be used in a classroom/laboratory setting to give students hands on experience in the integration of hardware and software equipment used for GPS research. By using GPS, students will be able to predict future trajectories of space vehicles. Students will visualize airplane and shuttle launch trajectories.

5. Acknowledgements

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6. References

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